Programming by Contract

CCC Level 1 (Summer)

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Motivation

- Remember how we are encouraged to spot bugs early on?

- *Programming by Contract* ensures that bugs show up **EARLY** and **FAIL HARD**

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Origins

- Also known as:
  - *"Design by Contract"*
  - *"Design by Contract Programming"*
  - *"Contract Programming"*

- Pioneered by Bertrand Meyer in the Eiffel Programming Language

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Main Idea

When a method of a class in OOP provides a certain functionality, it:

- Expects a certain condition to be guaranteed on entry by any client module that calls it (the method's precondition)
- Guarantees a certain property on exit (the method's postcondition)
- Maintains a certain property, assumed on entry and guaranteed on exit (the class invariant)

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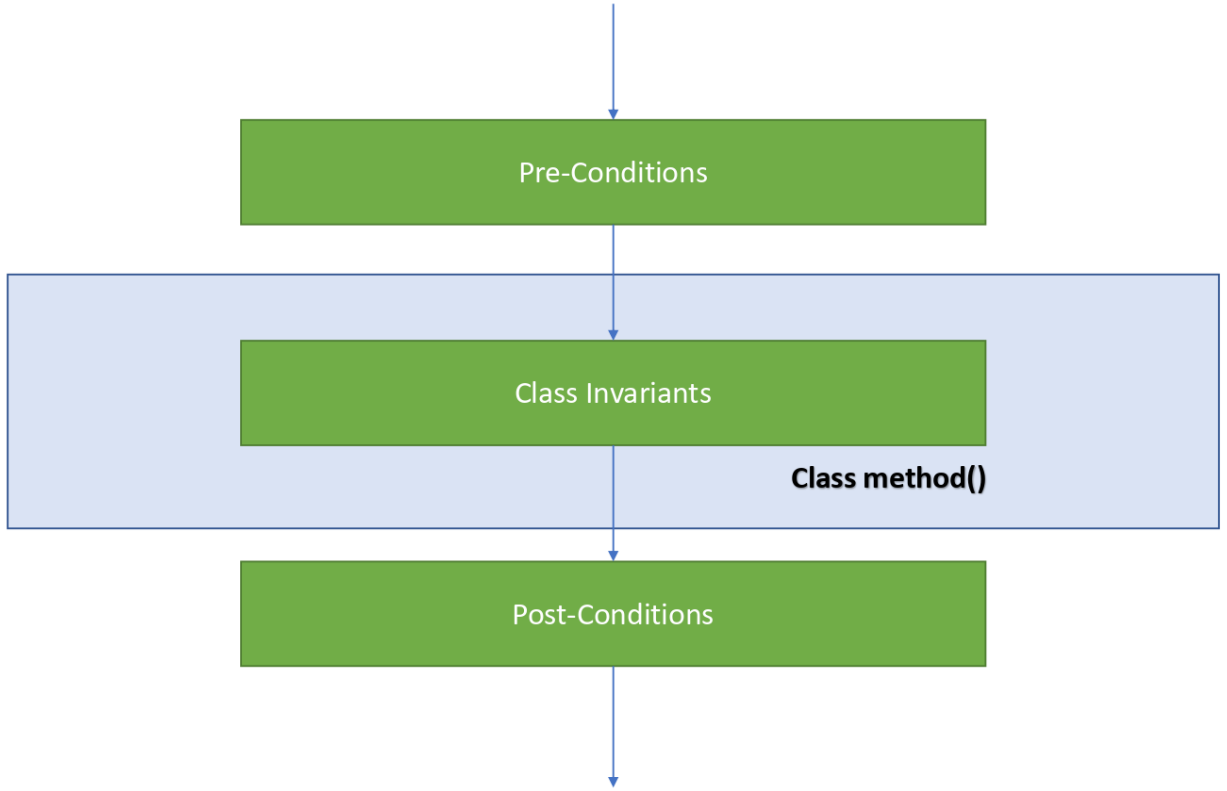
Main Idea



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Method Contracts

A method contract may contain the following:

- Input types, values, and their meanings
- Return types, values, and their meanings
- Errors and exception conditions
- Side effects
- Preconditions, Postconditions, and Invariants
- Performance guarantees

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Subclasses

- May weaken (and must not strengthen) preconditions
- May strengthen (and must not weaken) postconditions and invariants

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Assertions

- The **`assert`** keyword is available in Python, Java, and C++ to enforce conditions
- These **`assert`** statements are enabled during software testing
- Assert statements are disabled in production because they must *fail gracefully*, not *hard*

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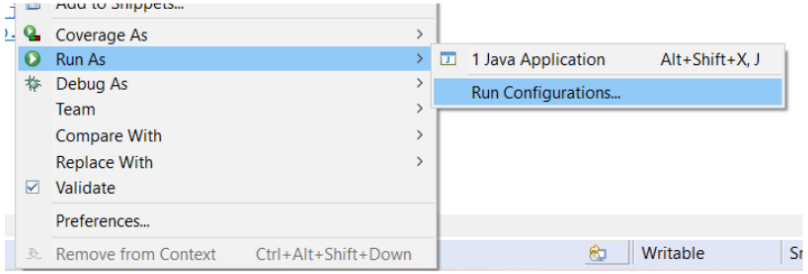
Assertions

- In Python and C++, **`assert()`** is enabled by default
- To enable **`assert()`** in Java, use **`–ea`** flag



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Assertions

For disabling the **`assert()`**:

- **Python**: Use **–O** flag
- **Java**: Use `–disableassertions` flag
- **C++**: Use `#define NDEBUG`
- **Alternative for C++**: Use `–DNDEBUG` flag with `g++`

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In-class Exercise 1

- Given the following code, please write assertion statements to determine the preconditions, postconditions, and invariants of the `car()` and `cdr()` methods.

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In-class Exercise 1

```python
def car(my_list):
  return my_list[0]

def cdr(my_list):
  return my_list[1:]

a = [2, 4, 6, 8, 10]
print(car(a))
print(cdr(a))
```

Output: 2

[4, 6, 8, 10]

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In-class Exercise 1

```java
import java.util.Arrays;

public class Lisp {

  public static int car(int[] arr) {
    return arr[0];
  }

  public static int[] cdr(int[] arr) {
    return Arrays.copyOfRange(arr, 1, arr.length);
  }
…
```

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In-class Exercise 1

```
…  …
12   public static void main(String[] args) {
13     int[] a = {2, 4, 6, 8, 10};
14     System.out.println(car(a));
15     System.out.println(Arrays.toString(cdr(a)));
16   }
17 }
```

Output: 2

[4, 6, 8, 10]

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In-class Exercise 1 Solutions

```python
def car(my_list):
  k = len(my_list)
  assert k > 0                          # Precondition

  ret_val = my_list[0]
  assert k == len(my_list)              # Invariant

  # Postcondition
  assert isinstance(ret_val, type(my_list[0]))

  return ret_val
```

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In-class Exercise 1 Solutions

```python
def cdr(my_list):
  k = len(my_list)
  assert k > 0                          # Precondition

  ret_val = my_list[1:]
  assert k == len(my_list)              # Invariant

  assert k-1 == len(ret_val)            # Postcondition

  return ret_val
```

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In-class Exercise 1 Solutions

```java
public static int car(int[] arr) {
  int k = arr.length;
  assert(k > 0);                          // Precondition

  int retVal = arr[0];
  assert(k == arr.length);                // Invariant

  // Postcondition
  assert(((Object) arr[0]).getClass().isInstance(retVal));
  return retVal;
}
```

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In-class Exercise 1 Solutions

```java
public static int[] cdr(int[] arr) {
  int k = arr.length;
  assert(k > 0);                          // Precondition

  int[] retVal = Arrays.copyOfRange(arr, 1, arr.length);
  assert(k == arr.length);                // Invariant

  assert(k-1 == retVal.length);           // Postcondition
  return retVal;
}
```

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Assertion Errors

- Happen when either *precondition*, *postcondition*, or *invariant* are *violated*

```
public static void main(String[] args) {
    int[] a = {};
    System.out.println(cdr(a));
}
```

Problems @ Javadoc Declaration Search Console History

```
<terminated> Lisp [Java Application] C:\Program Files\Java\jdk-11.0.4\bin\javaw.exe (
Exception in thread "main" java.lang.AssertionError
        at Lisp.cdr(Lisp.java:26)
        at Lisp.main(Lisp.java:35)
```

```
26      assert(k > 0);      // Pre-condition
```

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Assertion Errors

- Happen when either *precondition*, *postcondition*, or *invariant* are *violated*

```
print(cdr([]))
```

```
lisp
C:\Users\kcheung\PycharmProjects\test2\venv\Scripts\python.exe C:/Users/kcheung/PycharmProjects/test2/lisp.py
Traceback (most recent call last):
  File "C:/Users/kcheung/PycharmProjects/test2/lisp.py", line 29, in <module>
    print(cdr([]))
  File "C:/Users/kcheung/PycharmProjects/test2/lisp.py", line 22, in cdr
    assert k > 0     # Pre-Condition
AssertionError
```

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Summary

- Programing by Contract is important because it enables the developer to see bugs and fix them early on
- It involves assertion statements that reflect the preconditions, postconditions and invariants of a method
- Assertion errors occur when either one of these three are violated

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END

Questions?

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